

COMBINED PRIVATE AI + SLM BUILD/LEARN ROADMAP

Learn while building: the 7-day Private Financial SLM + RAG MVP is the practical project, while the SLM/private-AI roadmap supplies the concepts and resources behind each implementation stage.

PROJECT FOUNDATION

Target architecture: CSV → validation → SQLite → SQL/RAG → SLM router → verified data → grounded answer. The broader private-AI architecture is private data → RAG → local SLM → workflow → controlled application.

The financial MVP uses SQLite as the source of truth and deterministic calculations, RAG for retrieval/context, and the SLM for classification, routing and explanation.

DAY 1 — DATA PIPELINE + ML FOUNDATIONS

BUILD: CSV upload, parsing, mapping, validation, normalization, duplicate detection and error reporting.

LEARN: Python for ML, basic ML concepts, tokens, embeddings and the role of structured data.

RESOURCES: Hugging Face LLM Course — <https://huggingface.co/learn/llm-course/chapter1/1>

RESOURCES: Hugging Face Smol Course — <https://huggingface.co/learn/smol-course/unit0/1>

OUTCOME: CSV → validated transaction objects. No AI yet.

DAY 2 — SQLITE + DATA FOUNDATIONS

BUILD: SQLite schema, CRUD, indexes, duplicate prevention and financial aggregation queries.

LEARN: Why exact financial calculations belong in deterministic code/SQL rather than relying on model generation; structured vs. unstructured retrieval.

RESOURCES: Python sqlite3 documentation — <https://docs.python.org/3/library/sqlite3.html>

RESOURCES: Hugging Face LLM Course — <https://huggingface.co/learn/llm-course/chapter1/1>

OUTCOME: CSV → validation → SQLite → reliable financial queries.

DAY 3 — LOCAL SLM + CATEGORIZATION

BUILD: Local SLM inference, categorization, structured output, confidence score, batch categorization and human review.

LEARN: SLMs, open-weight models, inference, instruction tuning, SFT, chat templates and evaluation.

RESOURCES: Hugging Face Smol Course — <https://huggingface.co/learn/smol-course/unit0/1>

RESOURCES: Smol Course GitHub — <https://github.com/huggingface/smol-course>

RESOURCES: SmolLM — <https://github.com/huggingface/smolLM>

RESOURCES: Ollama documentation — <https://docs.ollama.com/>

OUTCOME: Transactions → local SLM → suggested categories → human verification.

DAY 4 — RAG + PRIVATE KNOWLEDGE

BUILD: Convert financial records to searchable representations, chunk, embed, index and retrieve with metadata.

LEARN: Chunking, embeddings, vector stores, similarity search, metadata filtering, retrieval pipelines, RAG evaluation and RAG vs. fine-tuning.

RESOURCES: LangChain Retrieval/RAG guide — <https://docs.langchain.com/oss/python/langchain/retrieval>

RESOURCES: Chroma Getting Started — <https://docs.trychroma.com/docs/overview/getting-started>

RESOURCES: Qdrant Documentation — <https://qdrant.tech/documentation/>

RESOURCES: Hugging Face LLM Course — <https://huggingface.co/learn/llm-course/chapter1/1>

OUTCOME: Question → retrieval → relevant financial context.

DAY 5 — QUERY ROUTING + PEFT/FINE-TUNING

BUILD: SLM router choosing SQL, RAG or Hybrid. SQL handles exact calculations; RAG handles contextual questions; Hybrid combines deterministic results with retrieved context.

LEARN: PEFT, LoRA, adapters, SFT, training/evaluation splits and why fine-tuning should teach behavior rather than store private records.

RESOURCES: Hugging Face PEFT — <https://huggingface.co/docs/peft/index>

RESOURCES: Hugging Face TRL SFT Trainer — https://github.com/huggingface/trl/blob/main/docs/source/sft_trainer.md

RESOURCES: Hugging Face Smol Course — <https://huggingface.co/learn/smol-course/unit0/1>

OUTCOME: User question → router → SQL/RAG/Hybrid → verified data → SLM.

DAY 6 — UI + PRIVATE INFERENCE + WORKFLOW ENGINEERING

BUILD: Dashboard and assistant UI. Show the source behind each answer.

LEARN: Local serving, quantization, CPU/GPU inference, REST APIs, containers, structured outputs, validation, workflow orchestration and human-in-the-loop controls.

RESOURCES: llama.cpp — <https://github.com/ggml-org/llama.cpp>

RESOURCES: vLLM serving documentation — https://docs.vllm.ai/en/latest/serving/online_serving/openai_compatible_server/

RESOURCES: Ollama documentation — <https://docs.ollama.com/>

OUTCOME: Usable private AI application with visible grounding and auditability.

DAY 7 — EVALUATION + SECURITY/GOVERNANCE + POLISH

BUILD: Evaluation set for SQL, routing, categorization, retrieval and final-answer correctness. Add constrained SQL, scoped RAG, citations and protection against model failures.

LEARN: Authentication, RBAC, encryption, secrets, network isolation, data minimization, PII/PHI detection, audit logging, retention, prompt injection, output validation and AI governance.

RESOURCES: OWASP GenAI LLM Top 10 (2026) — <https://genai.owasp.org/resource/owasp-genai-llm-top-10-2026/>

RESOURCES: NIST AI RMF — <https://www.nist.gov/itl/ai-risk-management-framework>

RESOURCES: NIST Generative AI Profile —

<https://www.nist.gov/publications/artificial-intelligence-risk-management-framework-generative-artificial-intelligence>

RESOURCES: HHS HIPAA Security Rule — <https://www.hhs.gov/hipaa/for-professionals/security/index.html>

OUTCOME: Reliable, auditable MVP with a security/governance foundation.

AFTER THE 7-DAY MVP — ADVANCED PRIVATE SLM PATH

The original MVP intentionally excludes fine-tuning, cloud infrastructure, bank APIs, OCR, mobile, autonomous actions and complex agent architecture. Treat those as Version 2/production extensions.

Next sequence: fine-tune with LoRA → compare base vs. tuned model → quantize/optimize → containerize model serving → private networking → stronger IAM → monitoring/audit → private deployment → formal compliance/security review.

For production-style serving, vLLM provides an HTTP server with OpenAI-compatible APIs; its documentation also warns that API-key authentication alone does not protect every endpoint, making it useful as a concrete lesson in secure model serving.

RESOURCE MAP — WHAT EACH RESOURCE IS FOR

Hugging Face LLM Course — foundations, Transformers, tokenization, inference and LLM concepts.

Hugging Face Smol Course — practical small-model training, evaluation and specialization.

PEFT — parameter-efficient fine-tuning, especially LoRA/adapters.

TRL SFT Trainer — hands-on supervised fine-tuning.

llama.cpp — lightweight local inference/model serving.

Ollama — approachable local model execution for early experimentation.

LangChain Retrieval — RAG architecture and retrieval building blocks.

Chroma — self-hostable vector storage/retrieval; useful for the MVP.

Qdrant — deeper vector search, filtering, hybrid retrieval and production-oriented retrieval concepts.

vLLM — higher-performance model serving and API-based deployment.

OWASP GenAI LLM Top 10 — practical GenAI/LLM application security risks.

NIST AI RMF + GenAI Profile — AI risk management and governance.

HHS HIPAA Security Rule — authoritative background for safeguards around ePHI when healthcare becomes a target domain.

BUSINESS IDEA TO REMEMBER

“We build private AI systems that allow organizations to automate sensitive workflows without sending their data to external AI providers.”

Do not sell only “an SLM.” Sell a complete private AI workflow. Potential industries: healthcare, insurance, finance, legal, government, accounting, HR and manufacturing.

Potential architecture: customer environment → sensitive database/documents → access-controlled RAG → task-specific local SLM → AI workflow engine → human/application output.

The SLM is a component; the commercial value is secure automation of a useful business workflow while maintaining control over sensitive data.

Important caveat: local inference does not automatically make a system HIPAA-compliant or otherwise secure. The database, application, logs, backups, network, model server, access controls, encryption and operational processes also require appropriate security and governance.